

Propositional Logic Intro

Note 1

Proposition: A statement with a truth value; it is either true or false.

Propositions can be combined to form more complicated expressions, using the following operations:

Operators		Quantifiers	Implication operations	
$\wedge$	and	$\forall$	for all	Implication $P \implies Q$
$\vee$	or	$\exists$	there exists	Inverse $\neg P \implies \neg Q$
$\neg$	not			Converse $Q \implies P$
$\implies$	implies			Contrapositive $\neg Q \implies \neg P$
$\equiv$	equivalent to			

Further, for an implication $P \implies Q$ where P is the *hypothesis* and Q is the *conclusion*, it is useful to know that $P \implies Q \equiv \neg P \vee Q$. Additionally, observe that any implication is logically equivalent to its contrapositive. $\neg(P \implies Q) \equiv P \wedge \neg Q$

DeMorgan's Laws: The following identities can be helpful when simplifying expressions and distributing negations.

- $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
- $\neg(\forall x P(x)) \equiv \exists x(\neg P(x))$
- $\neg(\exists x P(x)) \equiv \forall x(\neg P(x))$

1 Propositional Practice

Note 1

Convert the following English sentences into propositional logic and the following propositions into English. State whether or not each statement is true with brief justification.

Recall that $\mathbb{R}$ is the set of reals, $\mathbb{Q}$ is the set of rationals, $\mathbb{Z}$ is the set of integers, and $\mathbb{N}$ is the set of natural numbers. The notation " $a \mid b$ ", read as " a divides b ", means that a is a divisor of b .

(a) There is a real number which is not rational.

$$\exists x \in \mathbb{R}, x \notin \mathbb{Q} \quad \text{true}$$

(b) All integers are natural numbers or are negative, but not both.

$$\forall x \in \mathbb{Z}, ((x \in \mathbb{N}) \vee (x < 0)) \wedge \neg((x \in \mathbb{N}) \wedge (x < 0)) \quad \text{true}$$

(c) If a natural number is divisible by 6, it is divisible by 2 or it is divisible by 3.

$$\forall x \in \mathbb{N}, 6|x \Rightarrow (2|x) \vee (3|x) \quad \text{true}$$

(d) $\neg(\forall x \in \mathbb{Q})(x \in \mathbb{Z})$

存在一个数是有理数但不是整数 true

(e) $(\forall x \in \mathbb{Z})(((2|x) \vee (3|x)) \Rightarrow (6|x))$

如果一个整数被2或3整除, 那么它被6整除 ~~true~~ false

(f) $(\forall x \in \mathbb{N})((x > 7) \Rightarrow ((\exists a, b \in \mathbb{N})(a + b = x)))$

对任意 >7 的自然数, 它可以被分解为2个自然数之和 true

2 Truth Tables

Note 1

Determine whether the following equivalences hold, by writing out truth tables. Clearly state whether or not each pair is equivalent.

(a) $P \wedge (Q \vee P) \equiv P \wedge Q$

P	Q	$P \wedge (Q \vee P)$	$P \wedge Q$
0	0	0	0
0	1	0	0
1	0	1	0
1	1	1	1

X

(b) $(P \vee Q) \wedge R \equiv (P \wedge R) \vee (Q \wedge R)$

P	Q	R	$(P \vee Q) \wedge R$	$(P \wedge R) \vee (Q \wedge R)$
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

✓

(c) $(P \wedge Q) \vee R \equiv (P \vee R) \wedge (Q \vee R)$

P	Q	R	$(P \wedge Q) \vee R$	$(P \vee R) \wedge (Q \vee R)$
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

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3 Implication

Note 0
Note 1

Which of the following implications are always true, regardless of P ? Give a counterexample for each false assertion (i.e. come up with a statement $P(x, y)$ that would make the implication false).

(a) $\forall x \forall y P(x, y) \implies \forall y \forall x P(x, y)$.



(b) $\forall x \exists y P(x, y) \implies \exists y \forall x P(x, y)$.

$P(x, y) = (x < y)$

任意 x 存在 $y \Rightarrow y$ 不一定相同
存在 y 任意 $x \Rightarrow y$ 相同

(c) $\exists x \forall y P(x, y) \implies \forall y \exists x P(x, y)$.



锁定了 x 的值